

sky130_rhbd_tt_1P8_25C.ccs Library

Cell Groups
AND2X1
AND3X1
AO3X1
AOA4X1
AOAI4X1
AOI3X1
BUFX1
DFFQNX1
DFFQX1
DFFRNQNX1
DFFSNQX1
DFFSNRNX1
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DLATCHN
DLATCH
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NAND3X1
NOR2X1
NOR3X1
OR2X1
OR3X1

TIEHI
TIELO
TMRDFFQNX1
TMRDFFQX1
TMRDFFSNQX1
VOTER3X1

AND2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT	OUTPUT
B	Y
0	0
1	1

Footprint

Cell Name	Area
AND2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	B	Y
AND2X1	0.01066	9.23651

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND2X1	0.00000	5.40143	5.40293

Delay Information

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	B->Y (FF)	0.05161	0.55017	7.54000

Power Information

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	B	-0.01415	-0.92321	-14.96310
	B	3.14821	3.20820	4.08200

AND3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
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Truth Table

INPUT		OUTPUT
B	C	Y
0	x	0
1	0	0
1	1	1

Footprint

Cell Name	Area
AND3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	B	C	Y
AND3X1	0.01027	0.01055	9.43249

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND3X1	0.00000	5.00667	5.00784

Delay Information

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	B->Y (FF)	0.05807	0.57123	7.71865
	C->Y (FF)	0.05585	0.58520	7.91621

Power Information

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	B	-0.01415	-0.93491	-15.28060
	B	2.54706	2.60458	3.47060
	C	-0.01415	-0.93491	-15.28060
	C	2.53597	2.58844	3.35768

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!C * !Y)	0.00000	0.00000	0.00000
	(!C * !Y)	2.22277	2.22268	2.22243

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!C * !Y)	0.00000	0.00000	0.00000
	(!C * !Y)	2.78444	2.78367	2.78354

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.22829	2.22760	2.22711

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.78386	2.78352	2.78379

AO3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT
B	C	Y
0	x	1
1	0	0
1	1	1

Footprint

Cell Name	Area
AO3X1	76.51360

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	B	C	Y
AO3X1	0.01052	0.01062	5.66246

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AO3X1	0.00000	0.63380	1.19622

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	B->Y (FR)	0.09231	0.55812	4.85405
	C->Y (RR)	0.05280	0.58670	6.39091

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	C->Y (FF)	0.08002	0.59070	6.12096

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	B	0.00000	0.00000	0.00000
	B	1.52618	1.58202	2.33669
	C	0.00000	0.00000	0.00000
	C	83750.50000	83751.60000	83765.30000

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	C	0.00000	0.00000	0.00000
	C	88901.80000	88894.70000	88903.30000

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(C * Y)	0.00000	0.00000	0.00000
	(C * Y)	83718.60000	83723.00000	83737.90000

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(C * Y)	0.00000	0.00000	0.00000
	(C * Y)	0.08057	0.13140	0.88262

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.02430	0.02422	0.02425

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	1.47608	1.47610	1.47615

AOA4X1

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Truth Table

INPUT			OUTPUT
B	C	D	Y
0	x	x	0
1	x	0	0
1	0	1	1
1	1	1	0

Footprint

Cell Name	Area
AOA4X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	B	C	D	Y
AOA4X1	0.01053	0.01056	0.01066	9.47400

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOA4X1	0.00000	6.06594	6.67655

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	C->Y (FR)	0.13369	0.93445	11.96830
	D->Y (RR)	0.06669	0.80818	11.27680

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	B->Y (FF)	0.13119	0.60882	6.40497
	C->Y (RF)	0.09077	0.62567	7.49972
	D->Y (FF)	0.05821	0.66713	8.87069

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	C	0.00000	0.00000	0.00000
	C	87244.40000	87240.10000	87244.00000
	D	0.00000	0.00000	0.00000
	D	87662.20000	87653.20000	87656.10000

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	B	-0.01415	-0.93738	-15.34790
	B	5.10428	5.15887	6.01320
	C	0.00000	0.00000	0.00000
	C	84161.50000	84159.90000	84163.50000
	D	0.00000	0.00000	0.00000
	D	87083.80000	87077.00000	87089.70000

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(C * !Y) + (!C * !D * !Y)$	0.00000	0.00000	0.00000
	$(C * !Y) + (!C * !D * !Y)$	83941.80000	83944.50000	83948.70000

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(C * !Y) + (!C * !D * !Y)$	0.00000	0.00000	0.00000
	$(C * !Y) + (!C * !D * !Y)$	5.10219	5.15423	5.88447

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * !D * !Y)	0.00000	0.00000	0.00000
	(B * !D * !Y)	80068.10000	80085.50000	80100.50000
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	1.98405	1.98365	1.98375

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * !D * !Y)	0.00000	0.00000	0.00000
	(B * !D * !Y)	89999.50000	90007.40000	90004.20000
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	5.01393	5.01368	5.01393

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y) + (!B * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y) + (!B * !Y)	82551.60000	82551.10000	82553.20000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y) + (!B * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y) + (!B * !Y)	80039.90000	80043.20000	80044.20000

AOAI4X1

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Truth Table

INPUT			OUTPUT
B	C	D	YN
0	x	x	1
1	x	0	1
1	0	1	0
1	1	1	1

Footprint

Cell Name	Area
AOAI4X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	B	C	D	YN
AOAI4X1	0.01051	0.01059	0.01064	5.69310

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOAI4X1	0.00000	0.59852	1.13255

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	B->YN (FR)	0.10136	0.57047	4.88048
	C->YN (RR)	0.06148	0.60218	6.48967
	D->YN (FR)	0.02971	0.77948	10.33030

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	C->YN (FF)	0.09744	0.81222	8.88000
	D->YN (RF)	0.03265	0.80527	10.62030

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	B	0.00000	0.00000	0.00000
	B	1.43375	1.48670	2.23803
	C	0.00000	0.00000	0.00000
	C	84970.70000	84970.50000	84970.20000
	D	0.00000	0.00000	0.00000
	D	87608.30000	87598.40000	87600.20000

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	C	0.00000	0.00000	0.00000
	C	87785.10000	87778.80000	87785.40000
	D	0.00000	0.00000	0.00000
	D	88361.50000	88355.50000	88372.20000

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	$(C * YN) + (!C * !D * YN)$	0.00000	0.00000	0.00000
	$(C * YN) + (!C * !D * YN)$	84701.20000	84705.50000	84709.00000

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	$(C * YN) + (!C * !D * YN)$	0.00000	0.00000	0.00000
	$(C * YN) + (!C * !D * YN)$	1.43401	1.48731	2.22753

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * !D * YN)	0.00000	0.00000	0.00000
	(B * !D * YN)	80958.60000	80971.60000	80994.70000
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.02679	0.02672	0.02676

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * !D * YN)	0.00000	0.00000	0.00000
	(B * !D * YN)	90694.10000	90694.90000	90698.60000
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	1.37452	1.37448	1.37458

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN) + (!B * YN)	0.00000	0.00000	0.00000
	(B * C * YN) + (!B * YN)	83384.20000	83389.50000	83385.90000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN) + (!B * YN)	0.00000	0.00000	0.00000
	(B * C * YN) + (!B * YN)	80955.10000	80951.70000	80955.30000

AOI3X1

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Truth Table

INPUT		OUTPUT
B	C	YN
0	x	0
1	0	1
1	1	0

Footprint

Cell Name	Area
AOI3X1	59.28640

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	B	C	YN
AOI3X1	0.01060	0.01059	9.40285

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOI3X1	0.00000	0.68075	1.28396

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	C->YN (FR)	0.04424	1.82358	28.81680

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	B->YN (FF)	0.06086	0.56910	7.63712
	C->YN (RF)	0.02358	0.73302	11.09040

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	C	0.00000	0.00000	0.00000
	C	89216.00000	89216.00000	89202.90000

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	B	-0.01415	-0.93315	-15.23260
	B	1.64311	1.70043	2.54165
	C	0.00000	0.00000	0.00000
	C	82915.10000	82942.70000	83262.20000

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(C * !YN)	0.00000	0.00000	0.00000
	(C * !YN)	82795.50000	82799.30000	82835.80000

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(C * !YN)	0.00000	0.00000	0.00000
	(C * !YN)	0.07552	0.12178	0.81285

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.01858	0.01856	0.01853

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	1.60993	1.60946	1.60953

BUFX1

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Footprint

Cell Name	Area
BUFX1	42.05920

Pin Capacitance Information Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
BUFX1	0.00000	6.07079	6.07079

DFFQNX1

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Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
CLK	D	QN
x	x	1

Footprint

Cell Name	Area
DFFQNX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	CLK	D	QN
DFFQNX1	0.02312	0.01055	5.42591

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQNX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	CLK->QN (-R)	0.15767	0.77026	7.17577

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	CLK->QN (-F)	0.17583	0.86974	7.85531

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25610	26.26510	26.61580

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13080	21.13380	21.34910

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	21.09730	21.09820	21.30850
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	26.21600	26.21760	26.41830

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	10.32730	10.33190	10.54020
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	10.11630	10.12300	10.35220
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	9.90603	9.91184	10.11400
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	11.32080	11.32930	11.55020

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	24.93560	24.93560	24.93790
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	21.09450	21.09980	21.31620
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29680	11.30220	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	25.56940	25.56900	25.56920
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	10.10430	10.11330	10.34910
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30980	11.31770	11.53790

DFFQX1

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Truth Table

INPUT		OUTPUT
CLK	D	Q
x	x	1

Footprint

Cell Name	Area
DFFQX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	CLK	D	Q
DFFQX1	0.02313	0.01057	5.42070

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (-R)	0.13053	0.76910	7.17777

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (-F)	0.20829	0.87765	7.84306

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13010	21.14130	21.49430

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25770	26.26370	26.46880

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	21.09710	21.09840	21.30850
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	26.21550	26.21660	26.41790

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	10.11820	10.12490	10.35450
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	10.32710	10.33330	10.54000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	11.31920	11.32820	11.54880
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.90796	9.91381	10.11610

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	21.09440	21.09980	21.31600
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	24.93550	24.93560	24.93790
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29370	11.29950	11.50830

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	10.10430	10.11330	10.34910
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	25.56940	25.56900	25.56920
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30780	11.31570	11.53250

DFFRNQNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
CLK	D	RN	QN
x	x	x	1

Footprint

Cell Name	Area
DFFRNQNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	RN	QN
DFFRNQNX1	0.02338	0.01013	0.03121	3.68547

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (-R)	0.16979	0.68143	5.12498
	RN->QN (FR)	0.04861	0.72350	8.48475

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (-F)	0.24683	1.01050	7.91075

Constraint Information

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	RN ()	(CLK * D)	0.12041	0.62378	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.09231	0.62378	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.06890	0.62378	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.06890	0.62378	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78410	23.79030	24.08500
	RN	24.23280	24.24160	24.51420

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51660	34.51750	34.70790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	34.32890	34.32580	34.51390
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	23.89340	23.89490	24.08350
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	23.90710	23.90750	24.10180
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	23.89160	23.89270	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * QN)	0.00000	0.00000	0.00000
	(D * RN * QN)	13.81010	13.81360	13.99940
	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	16.55560	16.55700	16.77150
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	9.21207	9.21546	9.40994
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	9.21868	9.22340	9.42132
	(!D * RN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !QN)	13.90050	13.90930	14.13170
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	9.21591	9.21900	9.41697

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	22.75390	22.75620	22.75800
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	34.32620	34.32840	34.52350
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	22.73940	22.74200	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	18.84270	18.84570	19.03820
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	9.85869	9.85934	9.85928

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	23.36900	23.36970	23.37000
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	16.53620	16.54280	16.76100
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	23.36070	23.36090	23.36120
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	13.88330	13.89180	14.10360
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	10.43050	10.42980	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	22.01820	22.01430	22.01630
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	22.01010	22.00790	22.00910
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	13.81870	13.82300	14.02080
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	9.22476	9.22432	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	23.85240	23.85390	23.85450
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	23.85330	23.85480	23.85550
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	10.54350	10.54980	10.74640
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	10.91460	10.91330	10.91360

DFFSNQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
CLK	D	SN	Q
x	x	x	1

Footprint

Cell Name	Area
DFFSNQX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	SN	Q
DFFSNQX1	0.02229	0.01015	0.02175	3.66982

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (-R)	0.12574	0.65006	5.03573
	SN->Q (FR)	0.04992	0.73087	8.56161

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (-F)	0.23626	0.98340	7.74574

Constraint Information

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.07124	0.62378	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07124	0.62378	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.07592	0.62378	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.07592	0.62378	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64090	19.65090	19.96210
	SN	18.99750	19.01220	19.26980

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36350	35.37020	35.57670

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	19.79670	19.80060	20.01980
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	19.79210	19.79550	20.01260
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	35.14000	35.14270	35.34260
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	10.63540	10.64990	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	9.56950	9.58220	9.85092
	(D * SN * !Q)	0.00000	0.00000	0.00000
	(D * SN * !Q)	13.00380	13.00610	13.24300
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	9.56007	9.57095	9.81652
	(!D * SN * Q)	0.00000	0.00000	0.00000
	(!D * SN * Q)	14.92350	14.93260	15.17980
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	17.39070	17.39720	17.60760
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	3.48748	3.49844	3.80064

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN * Q) + (!CLK * !SN * Q)	19.80220	19.80550	20.00770
	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	33.71200	33.71360	33.71360
	(CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !SN * Q)	19.79860	19.80190	20.00590
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	14.27430	14.27810	14.47990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q) + (!CLK * !SN * Q)$	9.55386	9.56305	9.77692
	$(CLK * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q)$	34.35890	34.35940	34.36010
	$(CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q)$	9.55916	9.56779	9.78784
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	19.42270	19.43310	19.64740

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * D * Q)$	18.32540	18.33040	18.33270
	$(CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(CLK * !D * Q)$	9.86736	9.87317	9.87298
	$(!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * Q)$	9.87675	9.87554	9.87851
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	14.41110	14.41280	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	19.55150	19.55540	19.55620
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	11.02560	11.02730	11.02780
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	11.02780	11.02740	11.02850
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	3.67730	3.68628	3.89154

DFFSNRNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT	
CLK	D	RN	SN	Q	QN
x	x	0	0	1	1
x	x	0	1	0	1
x	x	1	0	1	0
x	x	1	1	1	1

Footprint

Cell Name	Area
DFFSNRNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)	
	CLK	D	RN	SN	Q	QN
DFFSNRNX1	0.02233	0.01012	0.03429	0.02266	3.55003	3.57679

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (-R)	0.17439	0.69565	5.08914
	QN->Q (FR)	0.05835	0.71471	8.26726
	SN->Q (FR)	0.05002	0.72185	8.45665

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (-F)	0.26496	1.49433	13.33210
	QN->Q (RF)	0.08615	0.81886	8.72930
	RN->Q (FF)	0.14637	1.57136	16.98700
	SN->Q (RF)	0.07335	0.88445	9.84025

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (-R)	0.16749	0.66986	5.02463
	Q->QN (FR)	0.04701	0.70034	8.24992
	RN->QN (FR)	0.04974	0.72224	8.47312

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (-F)	0.26400	1.64871	15.00370
	Q->QN (RF)	0.06712	0.87140	9.81575
	RN->QN (RF)	0.26883	1.07742	10.29660
	SN->QN (FF)	0.13574	1.63622	17.89340

Constraint Information

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	SN (R)	0.02841	0.07555	0.48844
	setup	SN (R)	0.02877	0.04029	1.33801

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	SN (R)	(CLK * D)	0.00303	-0.01763	-0.10399
	hold	SN (R)	(CLK * !D)	0.00404	-0.02015	-0.10798
	hold	SN (R)	(!CLK * D)	0.02841	0.07555	0.47919
	hold	SN (R)	(!CLK * !D)	0.02841	0.07555	0.48844
	setup	SN (R)	(CLK * D)	0.00711	0.02822	0.50011
	setup	SN (R)	(CLK * !D)	0.00898	0.02770	0.59579
	setup	SN (R)	(!CLK * D)	0.02830	0.04029	1.33801
	setup	SN (R)	(!CLK * !D)	0.02877	0.04029	1.05522

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11338	0.62378	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.11104	0.62378	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.08295	0.62378	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.08529	0.62378	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	RN (R)	0.05545	0.12339	0.68042
	setup	RN (R)	0.02405	0.01757	0.55630

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	RN (R)	CLK	0.04041	0.10325	0.63802
	hold	RN (R)	(!CLK * D)	0.05545	0.12087	0.66391
	hold	RN (R)	(!CLK * !D)	0.05541	0.12339	0.68042
	setup	RN (R)	CLK	0.02405	0.01757	0.48917
	setup	RN (R)	(!CLK * D)	-0.01254	-0.02266	0.55630
	setup	RN (R)	(!CLK * !D)	-0.01650	-0.02518	0.09646

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08529	0.62378	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08529	0.62378	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.09465	0.62378	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.09465	0.62378	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33600	16.33750	16.43300
	SN	-0.00707	-0.26008	-2.87550
	SN	16.29930	16.30400	16.40190

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25330	16.25540	16.35680
	RN	-0.00707	-0.26008	-2.87550
	RN	16.53210	16.53760	16.64060
	SN	32.67570	32.67930	32.87880

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25320	16.25530	16.35660
	RN	-0.00707	-0.26125	-2.89718
	RN	16.53210	16.53710	16.63980

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33610	16.33760	16.43370
	RN	32.89150	32.89600	33.17100
	SN	-0.00707	-0.26125	-2.89718
	SN	16.29960	16.30440	16.40090

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	32.63390	32.63580	32.82970
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	32.63280	32.63480	32.82630
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	32.44770	32.45000	32.64070
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	3.97134	3.97214	4.11830
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	32.45680	32.45830	32.65360
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	18.13490	18.14500	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	15.81350	15.82400	16.06380
	(D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q * QN)	16.61230	16.61480	16.82550
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	15.80900	15.81900	16.03970
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	16.18940	16.19090	16.39260
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	0.08317	0.09068	0.29350
	(!D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q * !QN)	17.60060	17.60710	17.85600
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	16.19150	16.19620	16.40660
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	5.63992	5.64756	5.93703

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	32.63830	32.64060	32.83720
	$(CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q * QN)$	31.10940	31.11060	31.11060
	$(CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q * !QN)$	32.63720	32.63970	32.83790
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	31.10040	31.10100	31.10090
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	17.57310	17.57630	17.77380
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	17.19820	17.20060	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	15.79930	15.80650	16.01610
	$(CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q * QN)$	31.73500	31.73480	31.73490
	$(CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q * !QN)$	15.80320	15.80910	16.02450
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	31.72880	31.72850	31.72900
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	17.58810	17.59760	17.81410
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	17.79830	17.79770	17.79840

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * D * SN * !Q * QN)$	30.36170	30.36280	30.36250
	$(CLK * !D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !D * SN * !Q * QN)$	30.35800	30.35420	30.35860
	$(!CLK * D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q * QN)$	16.91740	16.91820	17.10820
	$(!CLK * !D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !D * SN * !Q * QN)$	16.50760	16.50530	16.50610

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	32.24150	32.24300	32.24320
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	32.24100	32.24430	32.24480
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	18.12470	18.13260	18.32360
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	18.30360	18.30220	18.30280

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	30.73490	30.74010	30.74070
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	16.86220	16.86600	16.86550
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	16.86900	16.86640	16.87170
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	17.58650	17.59030	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	31.98660	31.99650	31.99740
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	18.05790	18.06370	18.06460
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	18.06310	18.06430	18.06470
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	6.06905	6.07850	6.28575

DFFX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT	
CLK	D	Q	QN
x	x	1	1

Footprint

Cell Name	Area
DFFX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	CLK	D	Q	QN
DFFX1	0.02311	0.01055	5.42033	5.42584

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (-R)	0.13037	0.76799	7.19024
	QN->Q (FR)	0.05010	0.81762	10.45400

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (-F)	0.22638	1.61070	17.57770
	QN->Q (RF)	0.05530	0.78645	9.93421

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (-R)	0.15787	0.76925	7.20227
	Q->QN (FR)	0.04415	0.80417	10.43860

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (-F)	0.19575	1.70306	18.81930
	Q->QN (RF)	0.04657	0.81108	10.40220

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56510	10.56770	10.67500

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12880	13.13120	13.23750

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12810	13.13120	13.23580

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56550	10.56790	10.67660

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	21.09730	21.09820	21.30850
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	26.21600	26.21760	26.41820

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	10.11630	10.12300	10.35220
	(D * !Q * QN)	0.00000	0.00000	0.00000
	(D * !Q * QN)	10.32730	10.33190	10.54020
	(!D * Q * !QN)	0.00000	0.00000	0.00000
	(!D * Q * !QN)	11.32070	11.32930	11.55020
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	9.90602	9.91182	10.11400

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	21.09450	21.09980	21.31620
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	24.93550	24.93560	24.93790
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.29680	11.30220	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	10.10440	10.11330	10.34910
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	25.56930	25.56900	25.56920
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.30980	11.31770	11.53790

DLATCHN

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	GATE_N	Q
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
DLATCHN	179.87680

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE_N	Q
DLATCHN	0.02276	0.01036	2.88873

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCHN	0.00000	22.21240	28.71060

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (RR)	0.18673	0.92901	6.98398
	GATE_N->Q (-R)	0.23633	1.05386	8.01314

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (FF)	0.15910	0.52878	3.42270
	GATE_N->Q (-F)	0.17422	0.57150	3.50583

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	13.54570	13.56700	14.09900
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	13.29300	13.30470	13.53950

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	9.19465	9.21965	9.74499
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	7.78815	7.79923	8.08816

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	16.22000	16.23130	16.47220

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	11.66510	11.67890	11.91190

Passive power(pJ) for GATE_N rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	16.31750	16.32770	16.59530
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	11.95450	11.96540	12.20380

Passive power(pJ) for GATE_N falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	11.77280	11.78580	12.03900
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.99105	9.00550	9.25036

DLATCH

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	GATE	Q
x	0	1
0	1	0
1	1	1

Footprint

Cell Name	Area
DLATCH	162.64960

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE	Q
DLATCH	0.02279	0.02073	2.90667

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCH	0.00000	18.14490	24.27320

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (RR)	0.18615	0.92758	6.99747
	GATE->Q (-R)	0.18553	0.91408	6.90747

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (FF)	0.15764	0.52799	3.43223
	GATE->Q (-F)	0.11822	0.44507	2.63795

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	14.12930	14.15170	14.69430
	GATE	0.00000	0.00000	0.00000
	GATE	13.88820	13.89540	14.14810

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	9.55328	9.57816	10.10730
	GATE	0.00000	0.00000	0.00000
	GATE	8.12183	8.12607	8.41114

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	12.15540	12.16640	12.40740

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	8.01379	8.02737	8.26039

Passive power(pJ) for GATE rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.37400	12.38230	12.61050
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.32353	9.33247	9.55308

Passive power(pJ) for GATE falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.18780	12.20130	12.45500
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.23767	8.24740	8.47690

INVX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
INVX1	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
INVX1	-

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
INVX1	0.00000	0.00191	0.00191

MUX2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A0	A1	S	Y
0	0	x	0
x	1	0	1
0	1	1	0
1	0	0	0
1	x	1	1

Footprint

Cell Name	Area
MUX2X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A0	A1	S	Y
MUX2X1	0.01056	0.01054	0.02177	5.42552

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
MUX2X1	0.00000	13.86620	17.36010

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
MUX2X1	A0->Y (RR)	-	0.07867	0.65088	6.88313
	A1->Y (RR)	-	0.07174	0.64161	6.84105
	S->Y (RR)	(A0 * !A1)	0.08261	0.63663	6.76020
	S->Y (FR)	(!A0 * A1)	0.11467	0.71920	7.19830

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
MUX2X1	A0->Y (FF)	-	0.07805	0.76794	8.29389
	A1->Y (FF)	-	0.07183	0.74250	8.18526
	S->Y (FF)	(A0 * !A1)	0.08054	0.76659	8.25537
	S->Y (RF)	(!A0 * A1)	0.10323	0.68517	6.75608

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
MUX2X1	A0	-	0.00000	0.00000	0.00000
	A0	-	9.83642	9.84536	10.09650
	A1	-	0.00000	0.00000	0.00000
	A1	-	5.46024	5.47007	5.71171
	S	(A0 * !A1)	0.00000	0.00000	0.00000
	S	(A0 * !A1)	10.13200	10.15450	10.69710
	S	(!A0 * A1)	0.00000	0.00000	0.00000
	S	(!A0 * A1)	5.71773	5.73163	5.96212

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
MUX2X1	A0	-	0.00000	0.00000	0.00000
	A0	-	9.85138	9.86439	10.08950
	A1	-	0.00000	0.00000	0.00000
	A1	-	5.47657	5.48974	5.72998
	S	(A0 * !A1)	0.00000	0.00000	0.00000
	S	(A0 * !A1)	5.14883	5.18392	5.74431
	S	(!A0 * A1)	0.00000	0.00000	0.00000
	S	(!A0 * A1)	9.55830	9.56879	9.80113

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A1 * !S * Y) + (!A1 * !S * !Y)	0.00000	0.00000	0.00000
	(A1 * !S * Y) + (!A1 * !S * !Y)	5.12297	5.12197	5.12366

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A1 * !S * Y) + (!A1 * !S * !Y)$	0.00000	0.00000	0.00000
	$(A1 * !S * Y) + (!A1 * !S * !Y)$	5.70563	5.70536	5.70573

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A0 * S * Y) + (!A0 * S * !Y)$	0.00000	0.00000	0.00000
	$(A0 * S * Y) + (!A0 * S * !Y)$	9.47388	9.47252	9.47443

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A0 * S * Y) + (!A0 * S * !Y)$	0.00000	0.00000	0.00000
	$(A0 * S * Y) + (!A0 * S * !Y)$	10.08620	10.08610	10.08610

Passive power(pJ) for S rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A0 * A1 * Y)$	0.00000	0.00000	0.00000
	$(A0 * A1 * Y)$	9.89281	9.91134	10.41400
	$(!A0 * !A1 * !Y)$	0.00000	0.00000	0.00000
	$(!A0 * !A1 * !Y)$	9.84683	9.85597	10.09490

Passive power(pJ) for S falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A0 * A1 * Y)	0.00000	0.00000	0.00000
	(A0 * A1 * Y)	5.47412	5.50232	5.99341
	(!A0 * !A1 * !Y)	0.00000	0.00000	0.00000
	(!A0 * !A1 * !Y)	5.42845	5.44170	5.68488

NAND2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT	OUTPUT
B	Y
0	1
1	0

Footprint

Cell Name	Area
NAND2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	B	Y
NAND2X1	0.01058	6.04081

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND2X1	0.00000	0.00114	0.00205

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	B->Y (FR)	0.02348	0.59829	8.07668

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	B->Y (FF)	0.00960	0.37225	5.09453

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	B	0.00000	0.00000	0.00000
	B	0.02064	0.03645	0.25536

NAND3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
B	C	Y
0	x	1
1	0	1
1	1	0

Footprint

Cell Name	Area
NAND3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	B	C	Y
NAND3X1	0.01028	0.01049	5.55329

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND3X1	0.00000	0.00074	0.00233

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	B->Y (FR)	0.02966	0.60893	7.98796
	C->Y (FR)	0.02616	0.62145	8.12502

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	B->Y (FF)	0.00941	0.37204	5.09449
	C->Y (FF)	0.00926	0.37197	5.09419

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	B	0.00000	0.00000	0.00000
	B	0.03036	0.04355	0.26237
	C	0.00000	0.00000	0.00000
	C	0.02118	0.03723	0.23096

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!C * Y)	0.00000	0.00000	0.00000
	(!C * Y)	-0.00204	-0.00207	-0.00206

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!C * Y)	0.00000	0.00000	0.00000
	(!C * Y)	0.01175	0.01110	0.01099

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.00049	0.00045	0.00043

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01145	0.01165	0.01167

NOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT	OUTPUT
B	Y
0	1
1	0

Footprint

Cell Name	Area
NOR2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	B	Y
NOR2X1	0.01054	8.77182

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR2X1	0.00000	3.02653	6.04921

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	B->Y (FR)	0.00867	0.37143	5.09380

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	B->Y (RF)	0.01469	0.42431	6.90250

Power Information

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	B	-0.01415	-0.89505	-14.21030
	B	2.73799	2.74627	2.83453

NOR3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Footprint

Cell Name	Area
NOR3X1	44.93040

Pin Capacitance Information Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR3X1	0.00000	0.00579	0.00579

OR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT	OUTPUT
B	Y
0	0
1	1

Footprint

Cell Name	Area
OR2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	B	Y
OR2X1	0.01073	5.66732

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR2X1	0.00000	5.40289	5.40294

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	B->Y (RR)	0.04420	0.38036	4.01549

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	B	0.00000	0.00000	0.00000
	B	2.94701	3.00462	3.86027

OR3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT
B	C	Y
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
OR3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	B	C	Y
OR3X1	0.01063	0.01059	5.56470

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR3X1	0.00000	2.84952	5.01932

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	B->Y (RR)	0.04842	0.38318	3.79539
	C->Y (RR)	0.04724	0.40887	4.19391

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	B	0.00000	0.00000	0.00000
	B	1.62046	1.67427	2.50858
	C	0.00000	0.00000	0.00000
	C	2.56312	2.61397	3.37565

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(C * Y)	0.00000	0.00000	0.00000
	(C * Y)	0.03957	0.03748	0.03691

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(C * Y)	0.00000	0.00000	0.00000
	(C * Y)	2.75698	2.75723	2.75709

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.01980	0.01979	0.01978

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	1.61031	1.61037	1.61035

TIEHI

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
TIEHI	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
TIEHI	12.24737

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TIEHI	0.00000	0.00000	0.00000

TIELO

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
TIELO	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	YN
TIELO	26.69489

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TIELO	0.00000	0.00000	0.00000

TMRDFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80, Temp
25.00

Truth Table

INPUT		OUTPUT
CLK	D	QN
x	x	1

Footprint

Cell Name	Area
TMRDFFQNX1	584.71600

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	CLK	D	QN
TMRDFFQNX1	0.07454	0.04011	2.70083

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQNX1	0.00000	71.39310	102.09600

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (RR)	0.37899	1.46068	10.48420

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (FF)	0.21246	0.35404	0.60428

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	76.32890	76.39910	77.72320

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	63371.80000	63388.20000	63406.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	65507.90000	66185.50000	66662.90000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	76.35840	76.41990	77.69390

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	103182.00000	104064.00000	100654.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	98401.60000	98513.00000	98726.30000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	98879.20000	98871.20000	98884.90000
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	88912.00000	88911.60000	89085.70000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	72662.50000	72662.10000	72663.10000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	72249.00000	72250.50000	72261.10000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71985.10000	71984.10000	71984.00000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	64002.90000	64003.30000	64004.30000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	64578.40000	64578.70000	64582.50000
	!CLK	0.00000	0.00000	0.00000
	!CLK	89970.30000	89973.70000	89975.80000

TMRDFFQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
CLK	D	Q
x	x	1

Footprint

Cell Name	Area
TMRDFFQX1	601.94318

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	CLK	D	Q
TMRDFFQX1	0.07443	0.04023	8.45426

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQX1	0.00000	77.76220	108.74900

Delay Information

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	CLK->Q (RF)	0.44943	1.08139	8.38492

Power Information

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	-0.01415	-0.87547	-13.69590
	CLK	80.66430	80.73090	82.01520

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	65806.20000	66448.50000	66896.70000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	63399.60000	63416.80000	63431.80000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	80.63330	80.69530	81.96440

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	98588.60000	98697.00000	98914.70000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	103425.00000	104171.00000	100551.00000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	88948.00000	88950.10000	89134.90000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	99585.00000	99571.40000	99573.60000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	72260.00000	72261.40000	72270.20000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	72293.20000	72291.50000	72294.80000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71608.90000	71606.50000	71609.50000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	64325.10000	64324.90000	64329.10000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	63862.60000	63863.80000	63864.60000
	!CLK	0.00000	0.00000	0.00000
	!CLK	87398.10000	87410.50000	87411.00000

TMRDFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80,
Temp 25.00

Truth Table

INPUT			OUTPUT
CLK	D	SN	Q
x	x	x	1

Footprint

Cell Name	Area
TMRDFFSNQX1	670.85199

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	SN	Q
TMRDFFSNQX1	0.07187	0.04118	0.07157	5.32224

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQX1	0.00000	85.26810	134.20100

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (-R)	0.24983	0.89108	7.10728
	SN->Q (FR)	0.16584	0.88762	8.02662

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (-F)	0.46612	0.97605	5.59773

Constraint Information

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.09699	0.62378	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.09699	0.62378	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.10168	0.62378	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.10168	0.62378	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	59.18130	59.19370	59.84400
	SN	58.46850	58.47770	58.78100

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	105.56500	105.57700	106.21500

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(D * SN) + (D * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * Q)$	59.21710	59.22720	59.86870
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	105.39800	105.40400	106.03400
	$(!D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * !SN * Q)$	33.00580	33.05030	33.83200

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(D * SN) + (D * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * Q)$	40.88360	40.89050	41.58610
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	53.50220	53.51140	54.13780
	$(!D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * !SN * Q)$	11.54440	11.57270	12.43390

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	100.05800	100.06500	100.06700
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	40.16520	40.17380	40.77120

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	101.96300	101.96600	101.97200
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	54.53600	54.56120	55.20190

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * Q) + (!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * Q) + (!CLK * D * Q)$	55.19510	55.19550	55.19960
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	44.14820	44.15670	44.73060

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * Q) + (!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * Q) + (!CLK * D * Q)$	58.53240	58.53710	58.54070
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	12.03750	12.06450	12.67380

VOTER3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
B	C	Y
0	x	0
1	x	1

Footprint

Cell Name	Area
VOTER3X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	B	C	Y
VOTER3X1	0.01987	0.01791	8.90702

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTER3X1	0.00000	4.53811	4.67509

Delay Information

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	B->Y (FF)	0.16764	0.86937	9.29050

Power Information

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	B	-0.01415	-0.90331	-14.42940
	B	1.03460	1.03519	1.12978

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(C * Y)	0.00000	0.00000	0.00000
	(C * Y)	52909.00000	52908.10000	52908.60000

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.15803	0.15812	0.15804
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	70298.10000	70356.30000	70361.30000

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	1.58673	1.58672	1.58671